

SWEEP rule on the KD-Toric code

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1 Counting Subtrees.

Claim 1.1. Let $G = (V, E)$ be a $\Delta + 1$ -regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. The number of subtrees rooted at r is at most α^k .

Proof. Denote by T_Δ and T_2 the infinity Δ and binary tree. There is an injection between the subtrees of G , rooted at v , with at most k edges, to the subtrees of T_Δ rooted at v , with at most k edges. Moreover, there is an injection from the subtrees of T_Δ rooted at v , with at most k edges to the subtrees of T_2 , rooted at v , with at most $k \log \Delta$ edges.

[COMMENT] In the second map, we think on replacing each of the vertex with $\Delta - 1$ -leaves binary tree, each leaf is associated with a outgoing edge. .

Thus, it's enough to bound the number of subtrees in T_2 rooted at v , with at most $k \log \Delta$ edges, that number is less than the number of binary tree with $k \log \Delta + 1$ leaves. For exact leaves number we get the $k \log \Delta$ Catalan number, which its limit is:

$$\sim \frac{4^{k \log \Delta}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}}$$

Therefore we can bound by:

$$\sum_{j=1}^k \frac{\Delta^{2j}}{(j \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta} \sum_{j=1}^k \frac{1}{j^{\frac{3}{2}}} \leq \zeta\left(\frac{3}{2}\right) \cdot \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

□

Now, from here we get an upper bound on the number of subgraphs that connected l vertices at the top level. We bound it from above by counting combination of rooted subtrees, such the vertices are their root and the overall edges number is summed up to k . Thus:

$$\leq \zeta\left(\frac{3}{2}\right) \binom{k-1}{l-1} \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$